

Video and Animation

Unit - 4

Video Signal Representation

- In conventional TV sets, the video signal is displayed using a cathode ray tube.
- An electron beam carries corresponding pattern information such as intensity in a viewed scene.
- Video signal representation includes three aspects:
 - the visual representation
 - Transmission
 - digitization

Visual Representation

- A key goal is to present the observer with as realistic as possible a representation of a scene.
- In order to achieve this goal, the television picture has to accurately convey the spatial and temporal content of the scene.
- Important measures taken for the purpose include:
 - Vertical Details and Viewing Distance
 - Horizontal Detail and Picture width
 - Total Content of a picture
 - Depth Perception
 - Luminance
 - Temporal aspects of illumination
 - Continuity of Motion
 - Flicker
 - Temporal aspect of bandwidth

Vertical Detail and the Viewing Distance

- Geometry of the field occupied by the television image is based on the ratio of picture width and height, called as aspect ratio ($W/L=4/3=1.33$)
- Viewing distance D determines the angle h subtended by the picture height.
- The angle is usually measured by the ration of the viewing distance to the picture height (D/H).
- Both the number of pixels per scanned line and the number of lines per frame vary; the actual numbers used being determined by the aspect ratio.
- The aspect ratio of television tubes is $4/3$ on which old PC monitors are based – and $16/9$ with widescreen television tubes.

Horizontal Detail and Picture Width

- The picture width normally used for television is $4/3$ times the picture height.
- The horizontal field of view can be determined using the aspect ratio.

Total Detail Content of the Image

- The vertical resolution is equal to the number of picture elements in the picture height, while the number of elements in the picture width is equal to the vertical resolution times the aspect ratio.
- The product of the picture's elements vertically and horizontally is the total number of picture elements in the image.

Depth Perception

- In natural vision, perception of the third spatial dimension, depth, depends primarily on the angular separation of the images received by the two eyes if the viewer.
- In the flat image television, a considerable degree of depth perception is inferred from the perspective appearance of the subject matter.
- The choice of the focal length of the camera lens and changes in depth of focus influence depth perception.

Luminance

- Color perception is achieved by three signals, proportional to the relative intensities of red, green, and blue light (RGB) present in each portion of the scene.
- Often a different signal division is used for transmission and storage: one brightness signal (luminance) and two color difference signals (chrominance).

Temporal aspect of illumination

- In contrast to continuous pressure waves of an acoustic signal, a discrete sequence of individual pictures can be perceived as a continuous sequence.
- Between frames, the light is cut off briefly. Two conditions must be met in order to represent a visual reality through motion pictures.
- First, the rate of repetition of the images must be high enough to ensure continuity of movements (smooth transition) from frame to frame.
- Second, the rate must be high enough that the continuity of perception is not disrupted by the dark intervals between pictures.

Continuity of Motion

- It is known that continuous motion is only perceived as such if the frame rate is higher than 15 frames per second.
- To make motion appear smooth, at least 30 frames per second must be used if the scene is filmed by a camera and not generated synthetically.
- Films recorded using only 24 frames per second often appear strange, especially when large objects move quickly and close to the viewer.
- NTSC(National Television Systems Committee) specified the frame rate was 30 frames/sec but has been changed to 29.97. PAL (Phase Alternating Line) adopted the frame repetition rate of 25 Hz, and the frame rate is 25 frames/sec.

Flicker

- If the refresh rate is too low, a periodic fluctuation of the perceived brightness can result. This is called the flicker effect.
- The minimum refresh rate to avoid flicker is 50Hz.
- Achieving continuous, flicker-free motion would thus require a high refresh rate.
- When the refresh rate of frames is smaller than the eye will notice the cutoff of light between the frames and there arises the flickering rate.

Temporal Aspect of Bandwidth

- The most important factor to determine at which bandwidth the video can be transmitted is its temporal specification.
- Temporal specification depends on the rate of the visual system to scan pixels, as well as on the human eye's scanning capabilities.
- In a HDTV (High Definition TV) device, a pixel can be scanned in less than a tenth of a millionth of a second.
- From the human visual perspective, the eye requires that a video frame to be scanned every $1/25$ second, this time is equivalent to the time during which human eye does not see the flicker effect.

Raw Video Size

- Consider PAL TV at 25 frames per second, sample at 352 x 288 with 16 bits per pixel.

Frame Size=Resolution Width×Resolution Height×Bits per Pixel

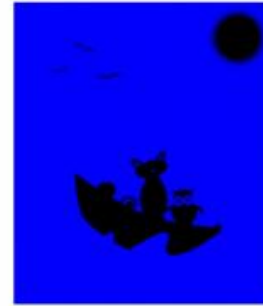
Data Rate=Frame Size×Frame Rate

Transmission

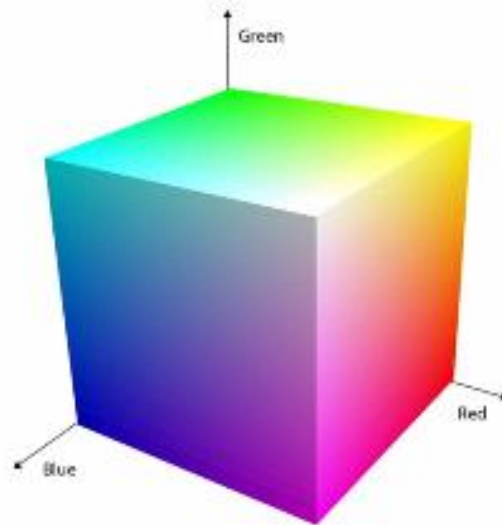
- Standards for transmission of video are: NTSC (National Television System Committee) and PAL (Phase Alternating Line).
- Approaches of color encoding are: RGB, YUV (luminance Y, two chrominance channels U and V), YIQ, Composite, etc.

RGB

- An RGB signal consists of separate signals for red, green, and blue.
- Every color can be encoded as a combination of these three primary colors using additive color mixing.
- The values R (for red), G (for green), and B (for blue), are normalized such that white results when $R+G+B = 1$ in the normalized representation.



RGB



YUV signal

- This model exploits the fact that the human eye is more sensitive to brightness i.e. luminance than to color information i.e. chrominance.
- Thus instead of separating the color component the brightness and the coloration of the image is separated.
- Thus luminance (Y) and two chrominance blue chrominance (U), and the red chrominance (V) are then used to represent the video content where

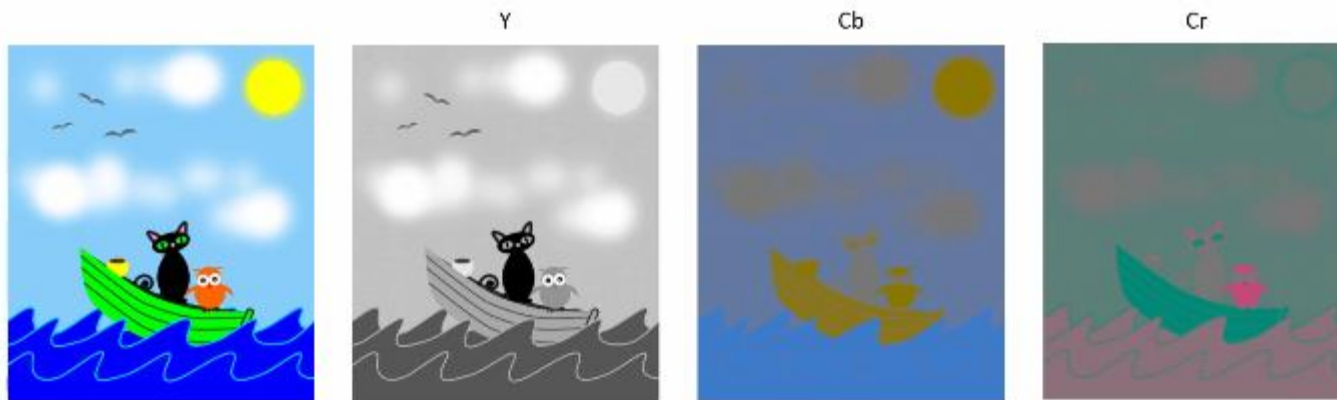
YUV signal

The YUV signal can be calculated as follows:

$$Y = 0.30R + 0.59G + 0.11B$$

$$U = (B - Y) \times 0.493$$

$$V = (R - Y) \times 0.877$$



YUV signal

- As the bandwidth for color broadcasts must be the same as monochrome, in order to fit the Y, U, and V signals in the same bandwidth, the three signals must be combined together for transmission.
- The resulting signal is then known as composite video signal.
- However due to the fact that chrominance is less informative the values of two chrominance is scaled down by certain scaling factor (this is done as a measure to save bandwidth).

YIQ signal

- This model is very much similar to YUV model and is used in NTSC system.
- The image information is broken down into three components a luminance component and two chrominance component given by following relation.

$$Y = 0.30R + 0.59G + 0.11B$$

$$I = 0.60R - 0.28G - 0.32B$$

$$Q = 0.21R - 0.52G + 0.31B$$

Composite Signal

- The alternative to component coding composes all information into one signal; consequently, the individual components (RGB, YUV, and YIQ) must be combined into one signal.
- The basic information consists of luminance information and chrominance difference signals.
- YUV signals tend to provide better quality because they keep color and brightness separate, reducing interference.

Digitization

- Digitization of video signals involves converting analog video signals into digital format.
- This process allows for improved quality, easier storage, and better transmission.
- This basically uses mathematical process like a series of step consisting of sampling and quantizing.
- Sampling is the process that actually digitizes the spatial position of the image while quantizing digitizes its color information.
- Sampling involves dividing the picture at $M \times N$ array of points while quantizing involves dividing the signal into a range of gray level values.
- Finally the digital motion video is created by digitizing the pictures temporally.

Computer Video Format

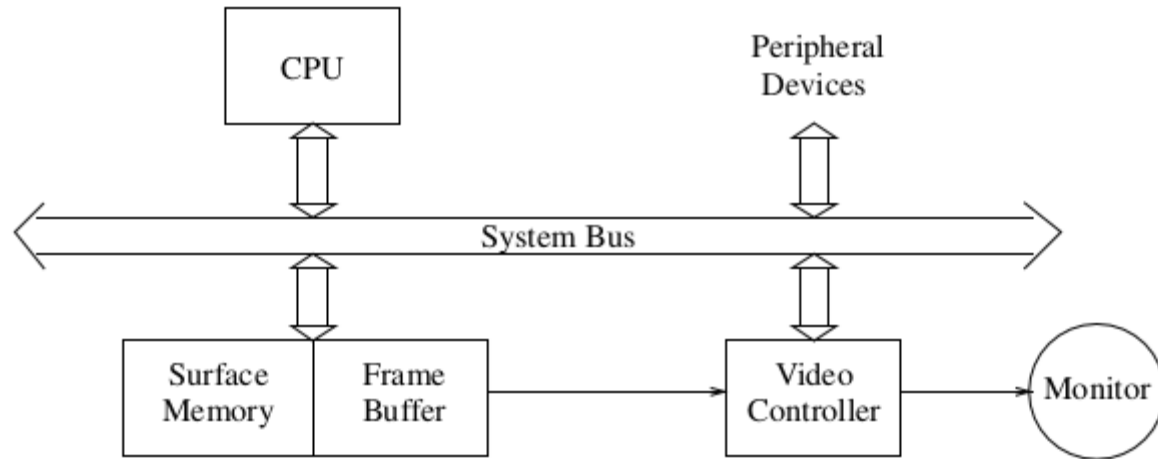
- In order to standardize the process of digitization the international body for television standards, the International Telecommunications Union – Radio communications Branch (ITU-R) defined a standard for the digitization of video pictures known as Recommendation CCIR-601
- In addition, a number of variants of this standard have been defined for use in other application domains such as television broadcasting, video telephony, and videoconferencing.
- The video digitizers differ in digital image resolution, quantization and frame rate.
- For e.g. VINO's resolution for NTSC is 640 by 480 with 8 bits/pixel and frame rate of 4 frames/second.

Computer Video Format

Computer video format depends on the input and output devices for the motion video medium

- Current video digitizers differ in digital image resolution, quantization and frame rate
- Most often used display is raster display
- The raster display architecture (as shown below)

Computer Video Format



4 Architecture of a raster display.

Computer Video Format

- Typically, the graphics program is used to create the high-level version of the image interactively and the Display controller part of the program interprets sequences of display commands and converts them into displayed objects by writing the appropriate pixel values into the refresh buffer.
- Normally the video controller is a hardware subsystem that reads the pixel values stored in the refresh buffer in time-synchronism with the scanning process and, for each set of pixel values, converts these into the equivalent set of red, green, and blue analog signal for output of the display.

Computer Video Format

- A range of color values are stored in a table and each pixel value is used as an address to a location within the table which contains the corresponding three color values. The table is known as color look up table.
- Some Video controller standards are given
 - Color Graphics Adapter (CGA)
 - Enhanced Graphics Adapter (EGA)
 - Video Graphics Array (VGA)
 - 8514/A Display Adapter Mode
 - Extended Graphics Array (XGA)
 - Super VGA (SVGA)

Computer Video Format

Color Graphics Adapter (CGA)

Resolution: 320*200

Color depth: 2 bits/pixel

Image size: 16 KB

No. of colors: 4

Storage Capacity: $320*200*4/8=16,000$ bytes

Enhanced Graphics Adapter (EGA)

Resolution: 640*480

Color depth: 4 bits/pixel

Image size: 112 KB

No. of colors: 16

Storage Capacity: $640*480*4/8=112,000$ bytes

Video Graphics Array (VGA)

Resolution: 640*480

Color depth: 8 bits/pixel

Image size: 307.2 KB

No. of colors: 256

Storage Capacity: $640*480*8/8=307,200$ bytes

8514/A Display Adapter Mode

Resolution: 1024*768

No. of colors: 256

Storage Capacity: $1024*768*8/8=786,432$ byte

Extended Graphics Array (XGA)

Resolution: 640*480 / 1024*768

Color depth: 65,000 colors / 256 colors

No. of colors: 256

Storage Capacity: $1024*768*8/8=786,432$ bytes

Super VGA (SVGA)

Resolution: 1024*768

Color depth: 24 bits/pixel

Image size: 2.35 MB

No. of colors: 2^{24}

Storage Capacity: $1024*768*24/8=2,359,296$ bytes

Computer-Based Animation

- Animation is the processes of making something look and feel live. Animation can broadly be categorized into two categories:

- **Motion Dynamics**

Induction of life by changing the position of the object

Ex: animation of a bouncing ball

- **Update Dynamics**

Induction of life by changing the shape, size and/or structure of the object

Ex: animation of flower blooming from bud

Computer-Based Animation

- Computer based animation used computer based graphical tools to provide visual effects.
- **Advantage of computer based animation** includes efficiency, time and simplicity.
- In several applications the video can be generated by a computer program rather than a video camera. This type of video content is normally referred to as **computer animation** or sometimes, because of the way it is generated, animated graphics.

Computer-Based Animation

- Computer based animation is an animation performed by a computer using graphical tools to provide visual effects.
- Animation covers all changes that have a visual effect. Visual effect can be of different nature.
- They might include:
 - Time-varying positions (motion dynamics),
 - Shape, color, transparency, structure and texture of an object (update dynamics), and
 - Changes in lighting, camera position, orientation and focus.

Basic Concepts

- Input Process
- Composition Stage
- In Between Process
- Changing Colors

Input process

- Before computer can be used, drawing must be digitized because key frames, in which the entities being animated are at extreme or characteristic positions, must be drawn.
- This can be done through optical scanning, tracing the drawings with a data tablet or producing the original drawings with a drawing program.
- Drawings may need to be post-processed (e.g. filtered) to clean up any glitches arising from the input process

Composition Stage

- In this stage, foreground and background figures are combined to generate individual frames for the final animation. This is done by image-composition techniques.
- The composition stage of computer-based animation is a critical phase where all separate elements foreground characters, background scenery, special effects and more are combined to create final frames of the animations

In-between process

- The animation of movement from one position to another requires the composition of frames with intermediate positions (intermediate frames) between key frames
- In computer-based animation, this in-between processing is done using interpolation (lerping).
- For example, if one uses lerping to calculate the intermediate positions of a ball that has been thrown in the air and uses only three key frames, as shown in Figure (a), then the resulting motion of the ball, depicted in Figure (b), is totally unrealistic.

In-between process



Figure 6-1 Linear interpolation of the motion of a ball: (a) key frames, (b) additional intermediate frames.

- Due to the disadvantages of lerping, splines are often used to smooth out interpolation between keyframes

Changing Colors

- For changing colors, computer based animators use CLUT (lut) in a frame buffer and the process of double buffering.
- The lut animation is generated by manipulating the lut.
- The simplest method is to cycle the colors in the lut, thus changing the colors of various pieces of the image.
- Using lut animation is faster than sending an entire new pixmap to the frame buffer for each frame.

Animation Language

- Animation languages fall into three categories:
 - Linear List notations
 - General purpose language
 - Graphical Language

Linear List Notations

- Each event in the animation is described by a starting and ending frame number and an action that is to take place (event).
- The actions typically take parameters, such as degree of rotation, axis of rotation, etc.
- E.g. 42, 53, B, ROTATE "PALM", 1, 30 means "between 42 and 53 rotate object called PALM about axis 1 by 30 degrees, determining the amount of rotation at each frame from table B.
- E.g.: Scefo (SCENE FOrmat)

General purpose language

- Another way to describe animation is to embed an animation capability within a general purpose programming language.
- The values of variables in the language can be used as parameters to the routines, which perform the animation
- E.g.: ASAS.

Ex:

```
(grasp my-cube); /* cube becomes the current object  
(cw 0.05); /* spin clockwise by a small amount  
(grasp camera); /* make the camera the current object  
(right 5); /*move right by 5 units
```


Graphical Language

- Graphical animation languages describe animation in a more visual way than textual animation languages.
- These languages are used for expressing, editing and comprehending the simultaneous changes taking place in an animation.
- Rather than writing out description of actions, the animator provides a picture of the action.
- Example.: GENESYSTM, S-Dynamics System.
- Graphical language describes animation in a more visual way.

Methods of controlling Animation

- Controlling animation is independent of the language used for describing it. Animation control mechanisms can employ different techniques.
 - Full explicit control
 - Procedural control
 - Constraint based system
 - Tracking live action
 - Kinematics and dynamics

Full explicit Control

- It is a simplest type of animation control.
- Animator provides a description of everything that occurs in the animation, either by specifying simple changes, such as scaling, translation, and rotation, or by providing key frame information and interpolation methods to use between key frames.
- This interpolation may be given explicitly or by direct manipulation with a mouse, joystick, data glove or other input device (in an interactive system). E.g. BBOP system

G. Stern. "Bbop—A system for 3D keyframe figure animation." In SIGGRAPH 83, pages 240 –243, New York, July 1983. Introduction to Computer Animation, Course Notes 7.

Procedural Control

- It is based on communication between various objects to determine their properties.
- It is a significant part of several other control mechanisms.
- In physically-based systems, the position of one object may influence the motion of another (eg: balls cannot pass through walls).
- In actor-based systems, the individual actors may pass their positions to other actors to affect the other actors' behaviors.

Constraint based system

- Many objects move in a manner determined by other objects with which they are in contact, and this compound motion may not be linear at all.
- Such motion can be modeled by constraints.
- Specifying an animated sequence using constraints is often much easier to do than using explicit control. E.g. Sutherland's Sketchpad

I. E. Sutherland. "Sketchpad: A man-machine graphical communication system." In SJCC, Spartan Books, Baltimore, MD, 1963.

Control by tracking live action

- Trajectories of objects in the course of an animation can be generated by tracking live action.
- Traditional animation uses rotoscoping. A film is made in which people/animals act out the parts of the characters in the animation.
- Then animators draw over the film, enhancing the background and replacing the human actors with their animated equivalents.

Control by tracking live action

- Another live-action technique is to attach some sort of indicator to key points on a person's body.
- By tracking the positions of the indicators, one can get location for corresponding key points in an animated model.
- Eg: data glove, which measures the position and orientation of the wearer's hand as well as the flexion and hyperextension of each finger point.



Kinematics and dynamics

- Kinematics refers to the position and velocity of points.
- E.g. a kinematic description of a scene might say, "The cube is at origin at time $t=0$. It moves with a constant acceleration in the direction $(1, 1, 5)$ thereafter.
- Dynamics takes into account physical laws that govern kinematics (Eg: Newton's laws of motion for large bodies etc.).

Kinematics and dynamics

- A particle moves with an acceleration proportional to the forces acting on it, and the proportionality constant is the mass of the particle.
- Thus, a dynamic description of a scene might be, "At time $t=0$ seconds, the cube is at position (0 meters, 100 meters, 0 meters). The cube has a mass of 100 grams. The force of gravity acts on the cube.
- Having the knowledge of kinematics and dynamics of the object at a point, we can determine the next point in motion.

Display of Animation

- Animated objects (which may consist of graphical primitives such as lines, polygons, and so on) must be scan-converted into their pixmap in frame buffer.
- To show a rotating object, we can scan-convert into the pixmap successive views from slightly different locations, one after another.
- This scan-conversion must be done at least 10 (preferably 15 to 20) times per sec to give a reasonably smooth effect.
- Hence, a new image must be created in no more than 100 milliseconds.

Display of animation

- To remove distracting effect (due to slow scan-conversion), double buffering is used.
- Let us assume that the two halves of the pixmap are *image0* and *image1*.

```
Load LUT to display values as background color
Scan-convert object into image0
Load LUT to display only image0
Repeat
    Scan-convert object into image1
    Load LUT to display only image1
    Rotate object data structure description
    Scan-convert object into image0
    Load LUT to display only image0
    Rotate object data structure description
Until (termination condition).
```

Transmission of Animation

- Animated objects can be represented symbolically using graphical objects or scan-converted pixmap images.
- Hence, the transmission of an animation may be performed using one of two approaches:

Symbolic Representation

- The symbolic representation (e.g., circle) of an animation's objects (e.g., ball) is transmitted together with the operations (e.g., roll the ball) performed on the object and at the receiver side the animation is displayed
- Since the byte size of such a representation is much smaller than a pixmap representation, the transmission time is short.
- The transmission rate (bits/sec or bytes/sec) of animated objects depends on:
 - The size of the symbolic representation structure, where the animated object is encoded,
 - The size of the structure, where the operation command is encoded, and
 - The number of animated objects and operation commands sent per second.

Pixmap Representation

- The pixmap representations of the animated objects are transmitted and displayed by the receiver.
- The transmission time is longer compared to the previous approach because of the size of the pixmap representation.
- However, the display time is shorter.
- The transmission rate of the animation is equal to the size of the pixmap representation of an animated object (graphical image) multiplied by the number of graphical images per second.